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Erasmus+ project ROBO STEAM ACADEMY
KA220-VET-7CF4F308

Lap Timer Troubleshooting and FAQ

Diagnose version 2 printing, fit and setup problems · Version 2.0

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About this guide

Start with the symptom you can observe. Change one thing at a time and record the result. Do not guess electrical connections or firmware behaviour.

1. Stop conditions

Disconnect power immediately if you see damaged insulation, smoke, unusual heat, a cracked PCB, a crushed cable, or a plug that cannot seat correctly.

> **Safety**
>
> Do not continue a test by holding damaged parts together. Repair or replace the part, then repeat the inspection from the beginning.

2. Printing problems

Symptom	Inspect	Safe next action
Base rocks	Warped floor or debris	Check the build-plate side and remove loose brim
Top will not seat	Locating lip, seam, or trapped cable	Test empty shells and inspect all four sides
Sensor hole is tight	Hole shrinkage or rough perimeter	Measure the test plate and change the CAD parameter
Screw splits coupon	Pilot too small or unsuitable screw	Try the next coupon hole; do not force it
Lug layers separate	Orientation or weak layer bonding	Stop tightening and revise the print setup
Rear opening has strings	Bridging or cleanup	Remove loose strings; do not enlarge blindly

Check: the observed fault and measurement are recorded before reprinting.

3. PCB fit problems

3.1 The board does not reach all four standoffs

1. Remove the board without bending it.
2. Check that its rear-left corner matches the drawing origin.
3. Measure the board outline.
4. Measure the four hole centres.
5. Compare the measurements with the engineering drawing.
6. Update the OpenSCAD parameters instead of forcing the board.

Check: the board can rest on all four standoffs without stress.

3.2 A locating pin is too large

Version 2 uses a 2.8 mm peg with a tapered lead-in. The scan suggests an approximately 3.4 mm board opening, but that edge was blurred. Measure the printed peg and physical PCB hole before changing the parameter.

4. Connector problems

4.1 The micro-USB plug does not seat fully

1. Disconnect the cable.
2. Inspect the molded plug body and strain relief.
3. Compare their largest dimensions with the opening.
4. Check the opening centre against the actual NodeMCU receptacle.
5. Mark the contact point on a photo.
6. Revise the opening width, height, centre, flare, or finger clearance.

Do not file the opening until the required change has been measured. A measured CAD revision gives a reproducible next version.

4.2 The DC barrel plug does not fit

Confirm both the opening size and centre. The current opening is provisional and may not match every HW-389-style clone.

5. Sensor problems

5.1 The sensor body touches the electronics

Stop assembly. Measure the actual highest electronics point and sensor body bottom. Version 2 assumes a 24 mm electronics envelope and gives 12 mm nominal clearance at a 45 mm axis. The first print showed that the inner sensor nut touched the NodeMCU at the lower version 1 height.

Possible measured revisions include increasing enclosure height, moving the sensor, or improving wire routing. Do not reduce the version 2 clearance based on the estimate alone.

5.2 The cable bends sharply

Measure the rigid sensor length and distance available behind it. Increase the box length or change the service-loop route for the next version.

5.3 The sensor is loose in the 19 mm hole

Use the retaining nuts correctly and record the printed hole size. Revise the panel-hole parameter only after measuring the sensor thread and printed test.

6. Closure problems

Symptom	Check	Response
Screw turns without tightening	Pilot damage or wrong screw	Stop and select a tested screw strategy
Boss cracks	Pilot too small or excessive torque	Reprint after changing the pilot or screw
Top shifts sideways	Locating lip fit	Inspect lip clearance and seam debris
Cable is pinched	Service-loop route	Disconnect power and reroute before closing
Seam remains open	Misalignment or collision	Remove the top and locate exact contact

7. Counting and calibration problems

The authoritative firmware, wiring, competition-platform protocol, and expected indications are not yet included. Use their verified troubleshooting steps when they are added.

For a mechanical check, confirm that the box has not moved, the sensor remains horizontal, the lens is clean, and no object blocks the sensing path.

8. Frequently asked questions

8.1 Why is the beam 45 mm high

The centred M18 sensor has a 9 mm radius. At a 45 mm axis its bottom is nominally 36 mm above the table. This leaves about 12 mm above the provisional 24 mm electronics envelope and prevents the inner retaining nut from touching the NodeMCU.

8.2 Can I print only one main part again

Yes. The base and top are separate STL files. Keep the same version label and record any parameter change that makes the reprinted part incompatible.

8.3 Can I use another NodeMCU or baseboard clone

Only after comparing its outline, holes, connector positions, and assembled height with the drawings. Product names alone do not guarantee identical geometry.

8.4 Can I power the system from any barrel adapter

No. Use only an adapter whose output voltage, polarity, current rating, connector, and safety approval match the verified electronics requirements.

8.5 Why are there three fit-test prints

The PCB coupon checks the measured hole pattern and board clearance. The other two pieces check the sensor opening and closure screw before the larger parts are printed.

9. Report a version 2 result

Include photographs, printer and material, slicer settings, measured dimensions, the exact symptom, and the smallest CAD change that may solve it. State whether the problem is repeatable.

About this document



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